

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 13

TEMPERATURE
FACTORS CONTROLLING
TEMPERATURE (PART - 3)



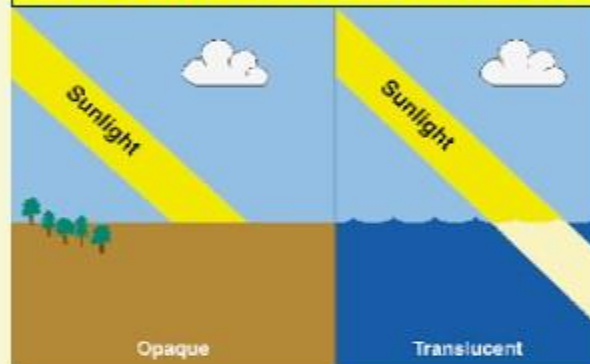
FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



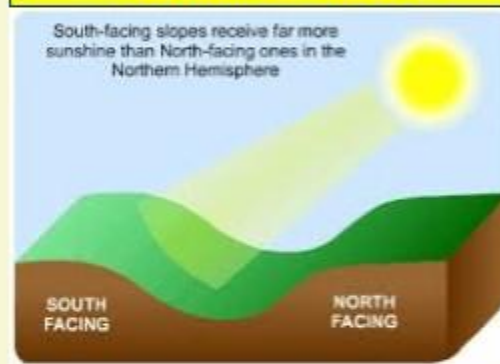
1. LATITUDES



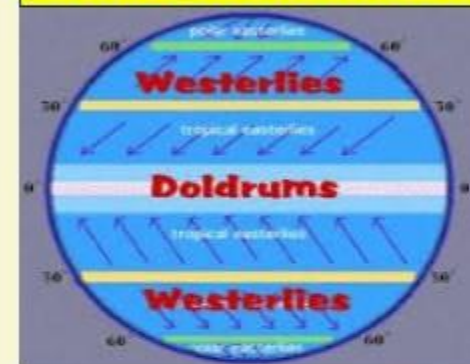
3. NATURE OF LAND & WATER



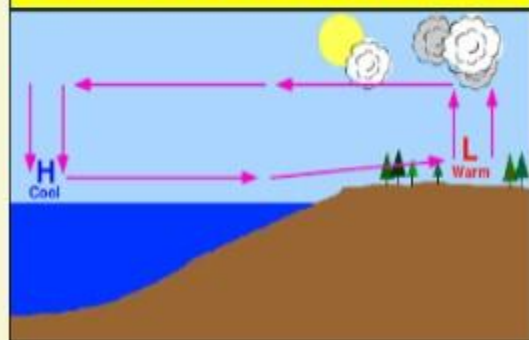
5. GROUND SLOPE



7. PREVAILING WINDS



2. DISTANCE FROM SEA



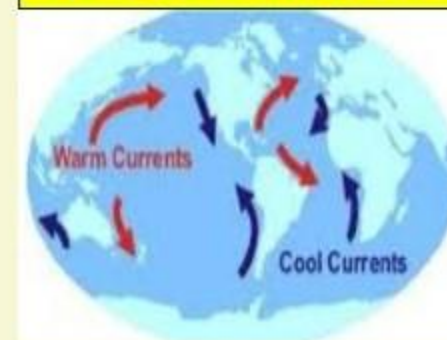
4. ALTITUDE



6. GROUND SURFACE



8. OCEAN CURRENTS





FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 7

NATURE OF LAND AND WATER

Continentality



DIFFERENTIAL HEATING

Sea heats up slowly and cools slowly.

Land heats up quickly and cools quickly.



1 The contrasting nature of land and water surfaces affects the **spatial and temporal distribution** of the temperature.

2 Land becomes warm and cold quickly than the water body.



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



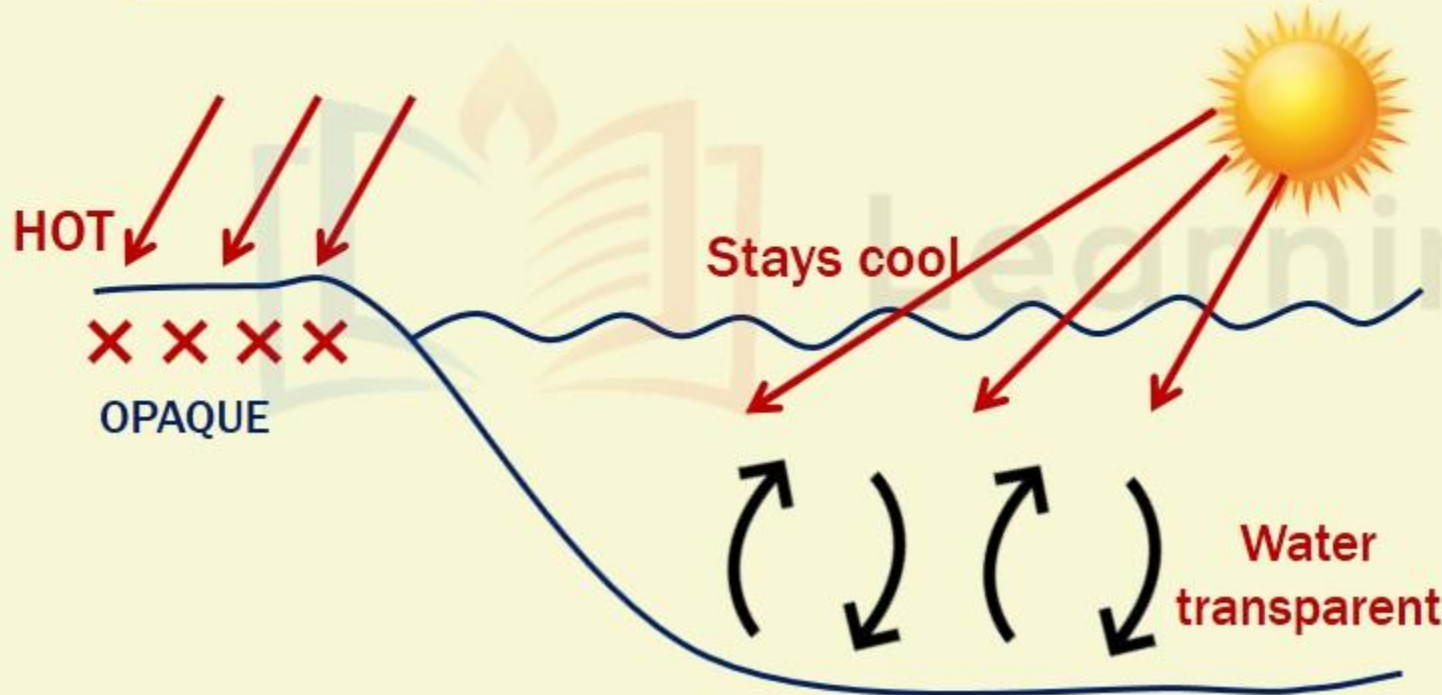
The following reasons explain the differential rate of heating and cooling of land and water:

FACTOR : 7

NATURE OF LAND AND WATER

REASON 1

LAND OPAQUE vs WATER TRANSPARENT



- 1** The land is **opaque** in nature whereas water is transparent.
- 2** Sun's rays penetrate to a depth of only 3 feet in land but they can penetrate to several metres in water with the heat getting spread to larger areas.
- 3** Hence the temperature of the ground surface may be larger than the oceanic surfaces.



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 7

NATURE OF LAND AND WATER

REASON 2



Ocean Currents

Caused due to gravity and wind friction.

Ocean Tides

Caused due to the Earth's rotation.

- 1 The **land is static but water is mobile**. Hence there is very slow redistribution of heat in the land.
- 2 Thus the land surface becomes very warm during the day and cold during the night
- 3 In seas, heat is redistributed **evenly by sea waves, ocean currents and tidal waves**.
- 4 All these extend the period of warming of water surface

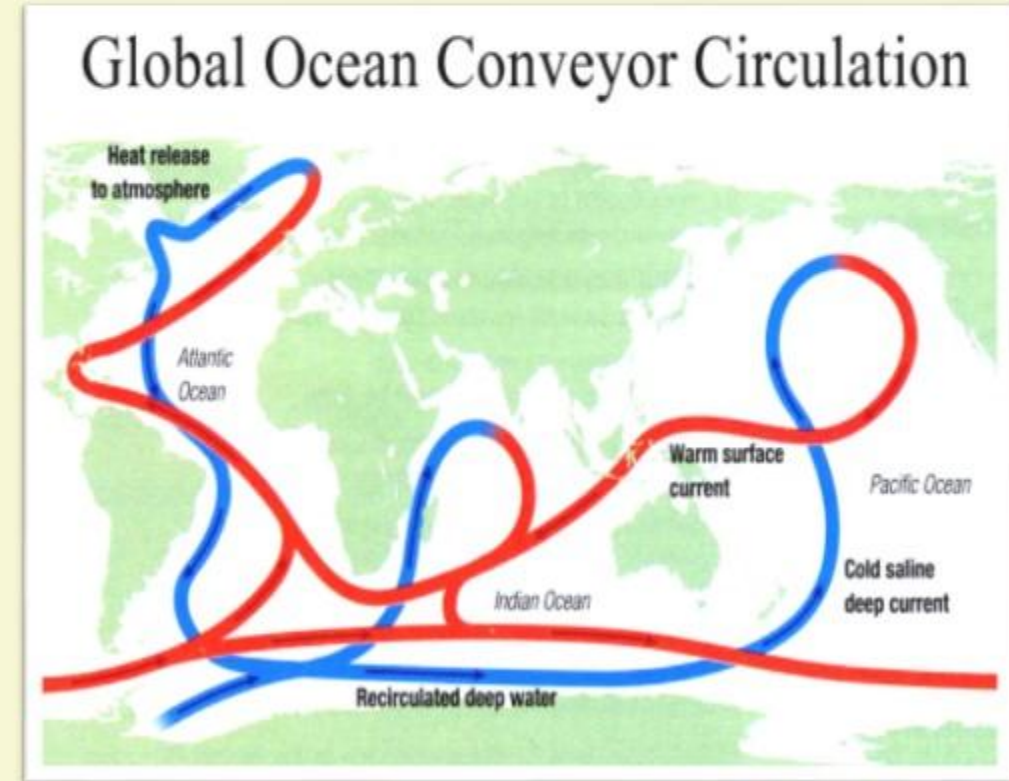
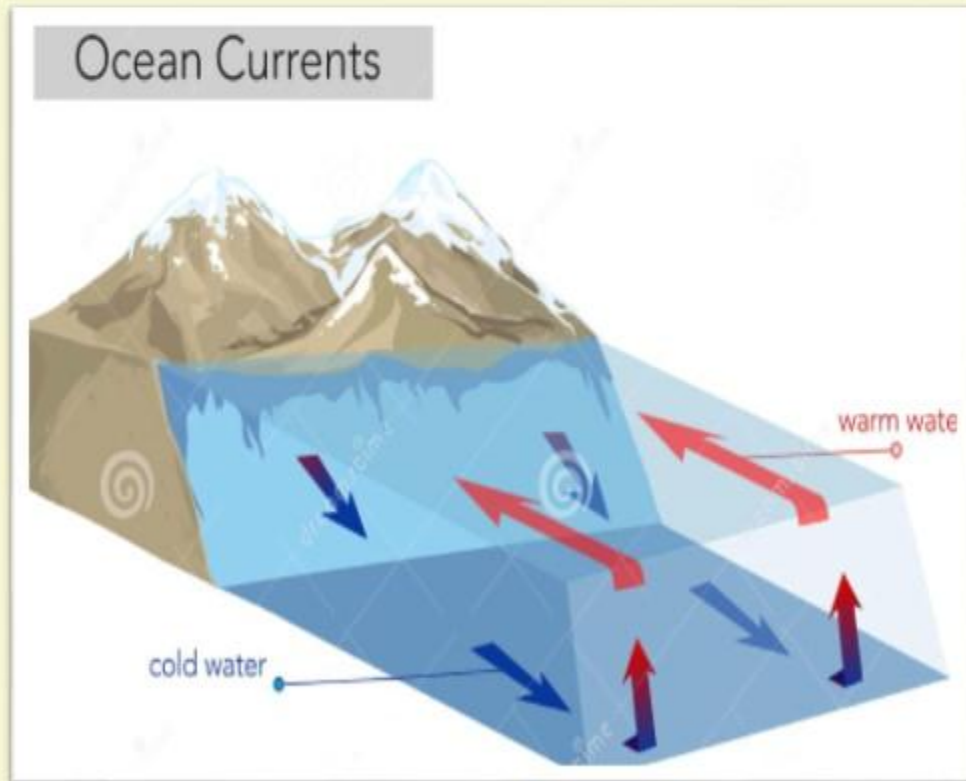


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 7

NATURE OF LAND AND WATER



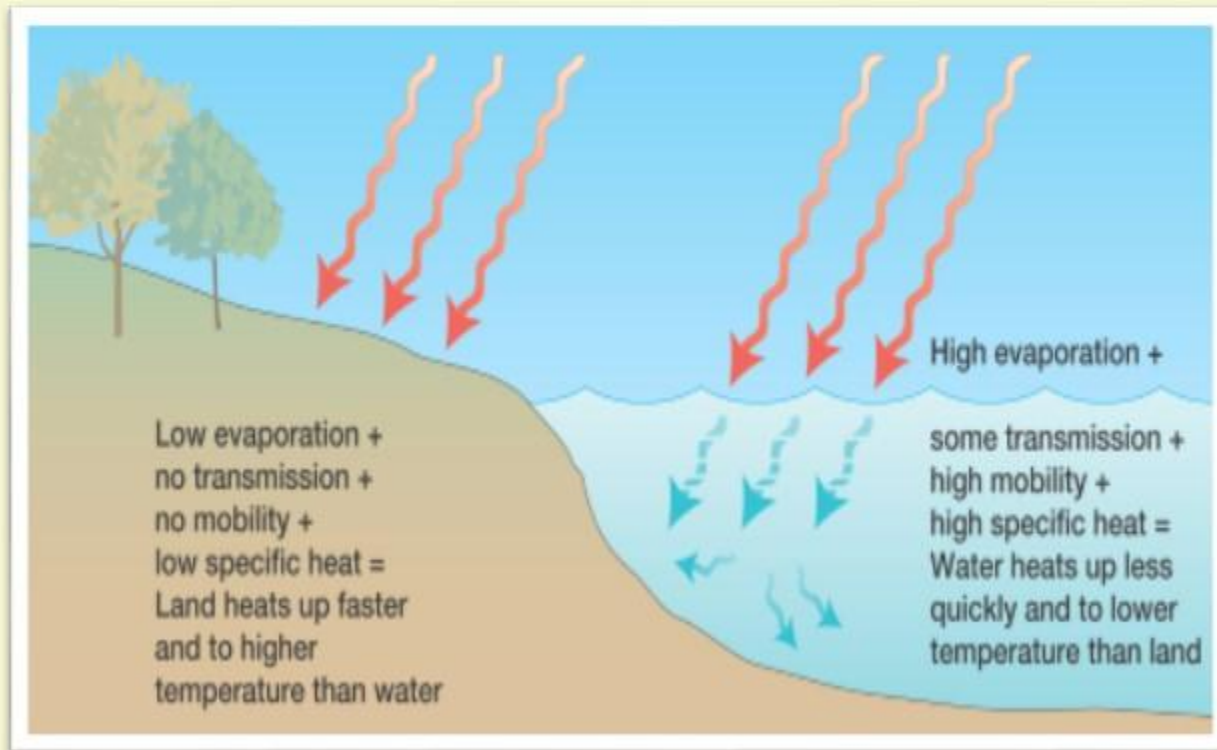


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 7

NATURE OF LAND AND WATER



REASON 3

There is **more evaporation** from the seas than the land and hence more heat is spent in the process



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 7

NATURE OF LAND AND WATER

Thinking About it...

On a hot summer day at the beach which will be hotter??

- A. The water in the Ocean
- B. The Land



WHAT IS SPECIFIC HEAT CAPACITY?

Specific Heat -

- Amount of heat it takes to raise 1g of a substance by 1°C

How easy or hard an object is to heat up – how fast does it absorb heat energy

- Metals have a low specific heat
- Water has a high specific heat





FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



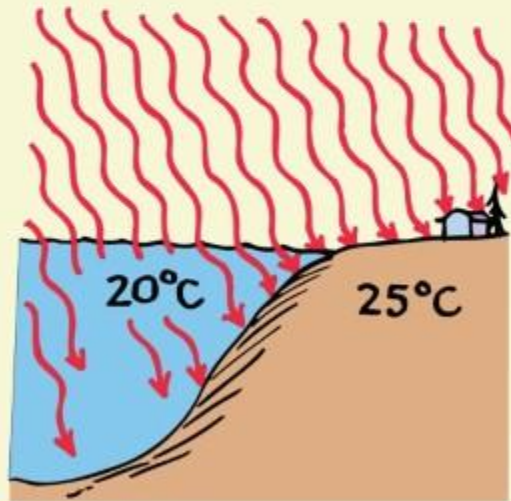
FACTOR : 7

NATURE OF LAND AND WATER

REASON 4

21.7 The High Specific Heat Capacity of Water

Water has a high specific heat and is transparent, so it takes more energy to heat up than land does.



- 1 Specific heat capacity of water is much greater than land.
- 2 It means that more heat is required to raise the temperature of one gram of water by 1°C than that of land
- 3 Therefore the **same amount of insolation** received by same mass of land and water **would increase the temperature of land more than the temperature of equal mass of water**

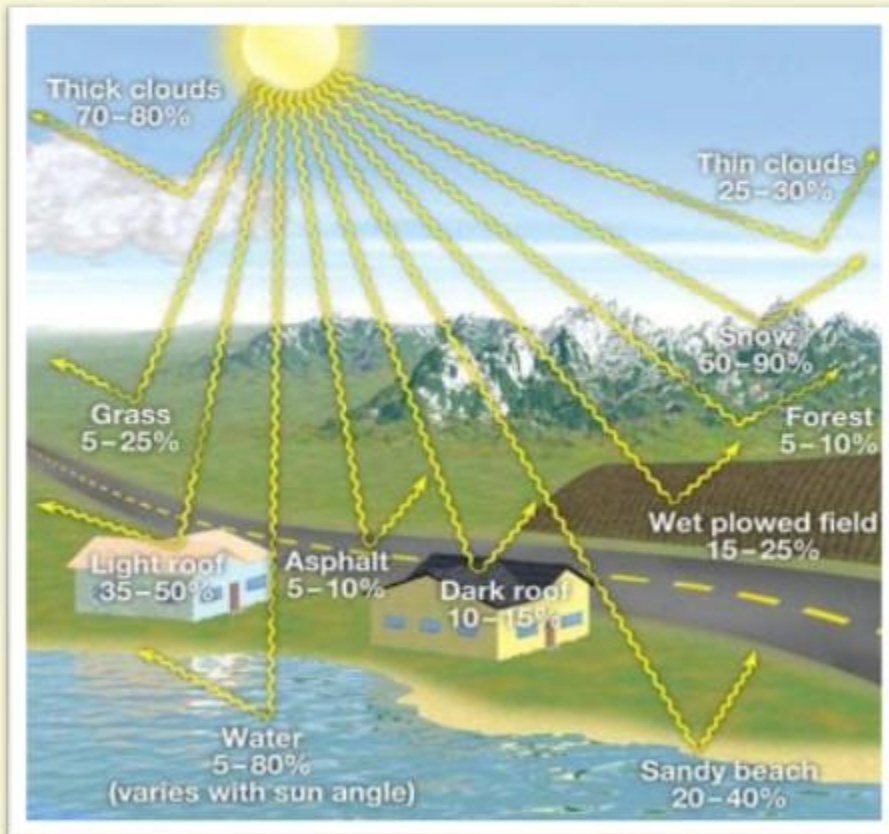


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 7

NATURE OF LAND AND WATER



REASON 5

The reflection (albedo) of incoming solar radiation from the oceanic water is far more than from the land surface.



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE

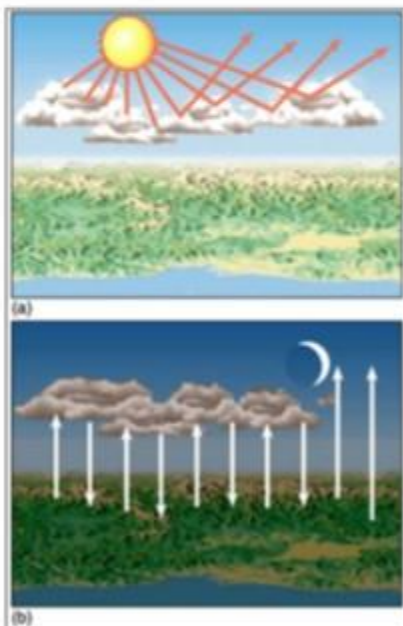


FACTOR : 7

NATURE OF LAND AND WATER

Cloud cover and albedo

Cloud cover can be important because many clouds have a high albedo.



REASON 6

- 1 The oceanic areas are generally more clouded than the land areas and hence they receive less insolation than land surface.
- 2 On the other hand, land surfaces receive more insolation at a faster rate because of less cloudiness and simultaneously lose more heat through outgoing terrestrial radiation very quickly.

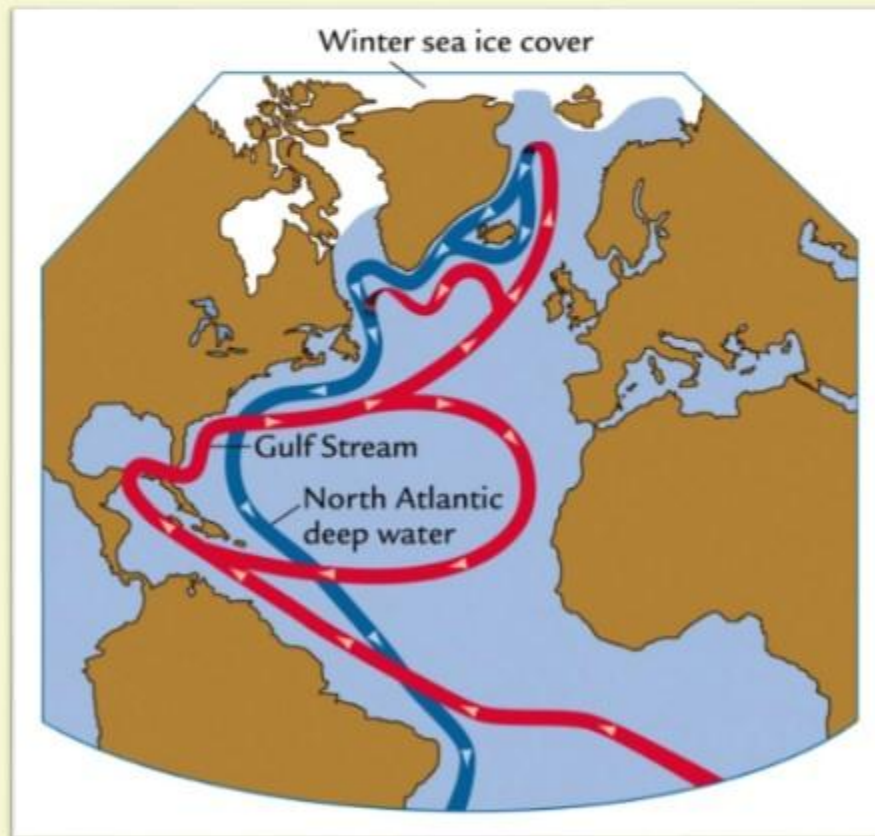


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE

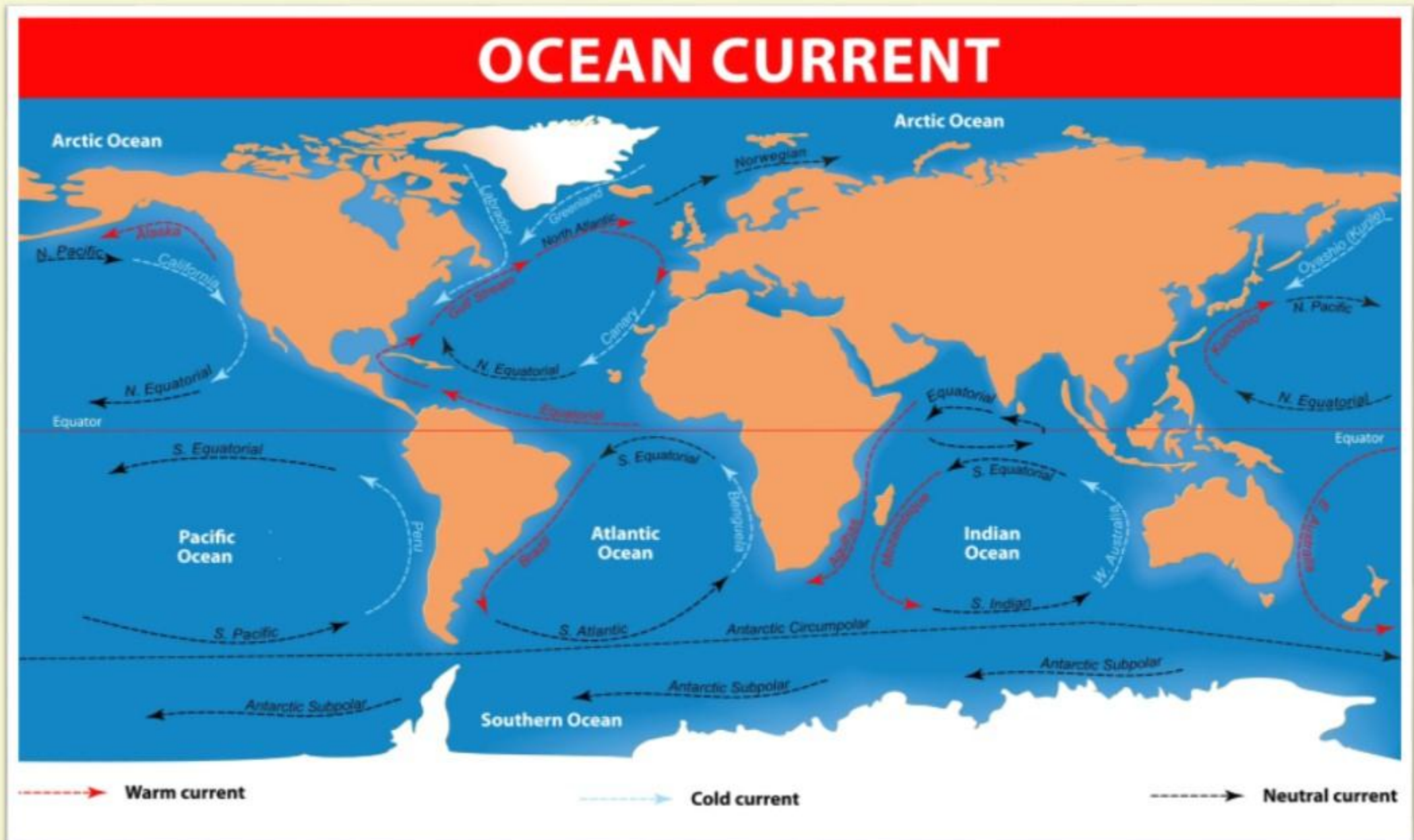


FACTOR : 8

OCEAN CURRENTS



- 1 An ocean current is the **movement of seawater in a definite direction** with a greater velocity generated by forces such as breaking waves, wind, temperature and salinity differences.
- 2 The places located on the coast where the **warm ocean currents flow record higher temperature** than the places located on the coast where the cold currents flow.



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